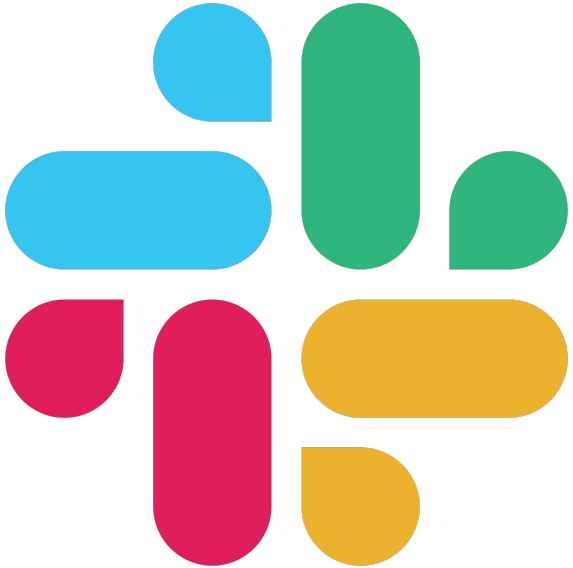
Bootstrapping, Bagging, and Ensembling

GENERAL ASSEMBLY

Agenda

- Recap of the Central Limit Theorem (CLT)
- The bootstrapping method
- **B**ootstrap **agg**regat**ing** (bagging) decision trees
- General ensemble methods for models
- Python!

Discuss: Remind me what the Central Limit Theorem says again?



The Central Limit Theorem (in a nutshell)

If you were to collect many repeated samples from a population, and each time recording the sample mean, that sample mean has a long run normal distribution.

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The Central Limit Theorem (in math)

This is often written:

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \longrightarrow N(0,1)$$

For large *n*, we can write the above (approximately) as:

$$\bar{X} \approx N\left(\mu, \sigma/\sqrt{n}\right)$$

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Pause and Ponder

The CLT is only concerned with means.

What would a CLT for other statistics look like?

Does the population *median* have a defined long-run distribution for an arbitrary iid set of *X*s?

Pause and Ponder

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What would a CLT for other statistics look like?

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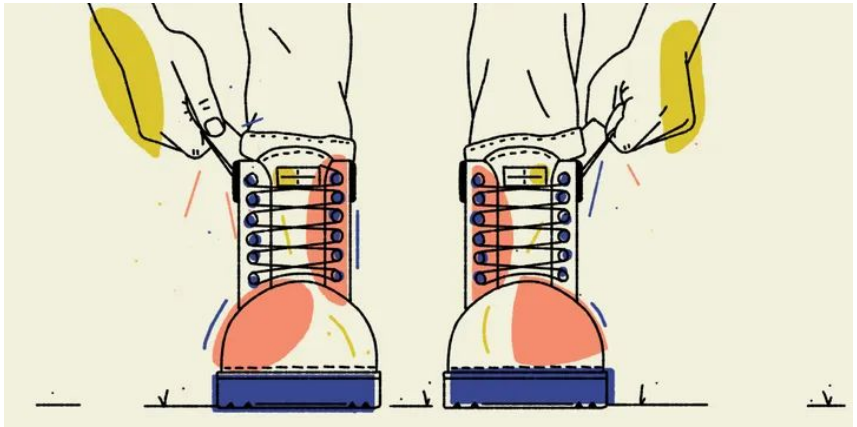
Actually, no!!!

Bootstrapping

The CLT fails for other statistics

What distribution does the population median have? The population 63rd percentile? The population standard deviation? The population *regression coefficient?*

We can approximate these things with **bootstrapping**.



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The Bootstrapping Technique

Bootstrapping requires 3 things:

- A sample size *n*
- A statistic *t* (e.g. the median)
- An arbitrary number *B* (usually a few hundred)

The Bootstrapping Technique

Given a sample size n, statistic t, and large number B:

1. Create a random sample from your data of size n **with replacement**
2. Compute the statistic t from this data and write it down
3. Perform steps 1-2 B times
4. You now have a set of sample statistics t_1, ..., t_B. This is your **bootstrapped distribution** and now represents your estimate of the distribution of t

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Let's try it out!



The Wisdom of the Crowds

Poll: How many jellybeans are in the jar?



The Wisdom of the Crowds

Individuals are dumb, but crowds are smart!

It's an actual statistical fact that, instead of relying on one "ok" model's opinion, you'll get better results if you aggregate many "ok" models together!

This act of **ensembling** via **bagging** reduces the overall variance of your model. Since decision trees tend to overfit, they're a prime opportunity for bagging.

While less effective, bagging "dumb" learners (ie, underfit models) also tends to improve model fit! Win-win!

Bagging

We can simulate the wisdom of the crowds to improve our “simple” decision trees. The method of **b**ootstrap **agg**regat**ing**, or **bagging**, works as follows:

1. Create a bootstrap sample of your data
2. Fit a model M (in our case a decision tree)
3. Repeat steps 1-2 B times (often $B = 500$, but it’s arbitrary)
4. You now have a set of models $M_1, \ldots, M_B$. Predictions are made as follows:

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Bagging Prediction

For a given sample, compute its predictions for each model M_i.

For **regression**, the final prediction is the **mean** of these predictions.

For **classification**, the final prediction is the **majority vote** of these predictions.

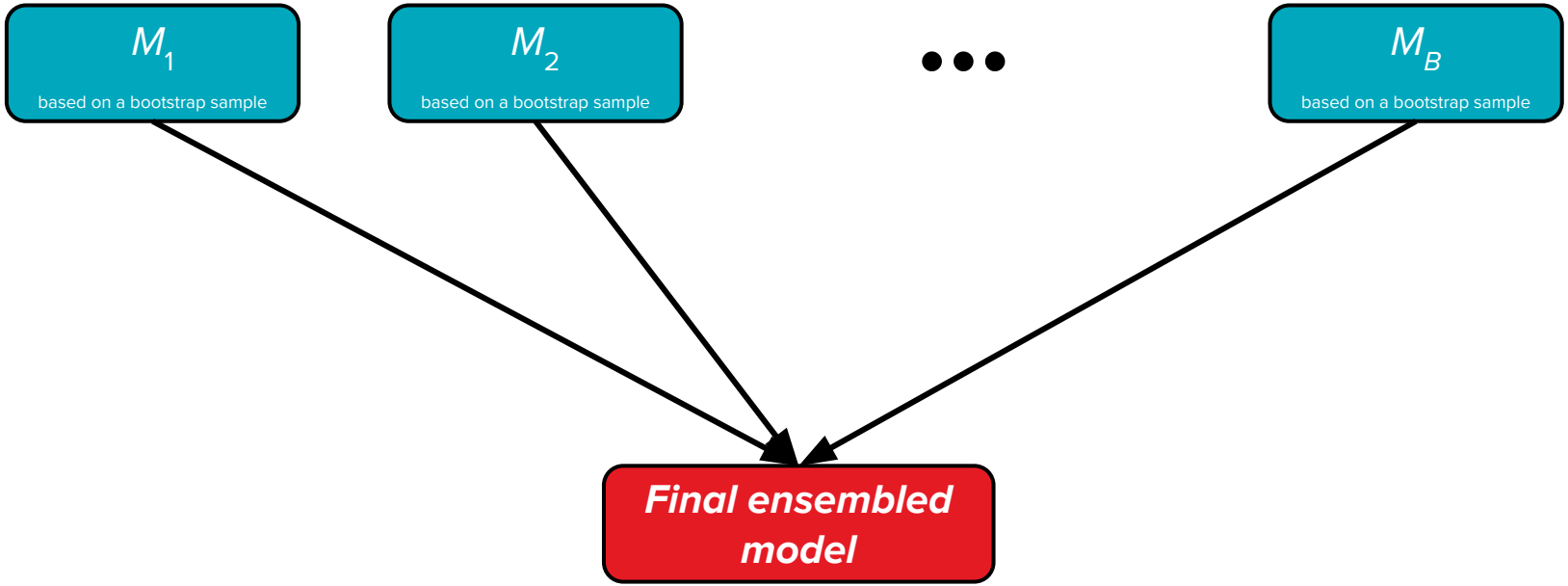
Ensembling

Ensemble Models

Bagging is one of many types of **ensembling** models together.

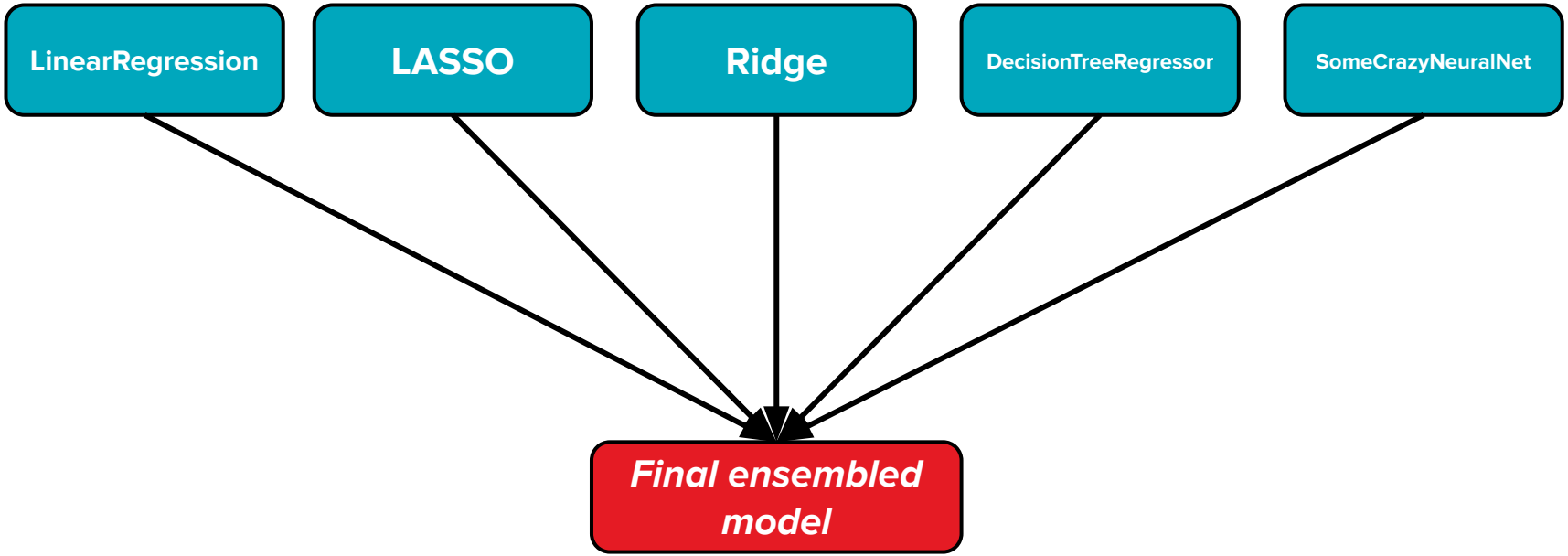
An **ensemble model** is one that is the result of combining the predictions of multiple other models.

Bagging is ensembling!



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We can also have arbitrary models ensembled together



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Let's try it out!

